

KEEGAN TU

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EDUCATION

Rutgers University

May 2026

B.S. Computer Science | GPA: 3.3

Relevant coursework: Database Systems, Data Structures, OOP (Java), Numerical Analysis, Data Science

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL (window functions, CTEs); learning TypeScript

Backend: FastAPI, REST APIs, JWT authentication, async programming, psycpg2

Frontend: React, HTML/CSS

Databases & Services: PostgreSQL, Redis

Tools: Git, GitHub, Linux, Bash, VS Code, Postman

PROJECTS

Workout Tracker | Python, FastAPI, PostgreSQL, React, JWT *workout-track.up.railway.app*

- Built a full-stack fitness tracking app with user authentication, workout logging, and progress analytics.
- Designed a normalized PostgreSQL schema and wrote raw SQL for all queries, including window functions for volume trends and RANK() over user totals for a leaderboard.
- Implemented JWT-based auth with password hashing (bcrypt), token expiration, and per-endpoint ownership checks to prevent unauthorized access to other users' data.
- Deployed the API and React frontend to Railway with a managed Postgres instance.

NLP-to-SQL Query System | Python, Qwen2.5-3B, PostgreSQL, SSH

- Connected a locally hosted 3B parameter language model to a live PostgreSQL database of HMDA mortgage data, enabling natural language queries against real financial records.
- Handled SSH tunneling, credential management, and query safety to prevent injection and destructive statements from generated SQL.

Stock Market Aggregator | Python, FastAPI, Redis, async httpx, React

- Built an async API aggregating live stock data and financial news from Alpha Vantage and News API.
- Added a Redis caching layer with TTL-based expiration to respect upstream rate limits and reduce response time on repeat queries.
- Used httpx to fan out concurrent requests across multiple external APIs from a single endpoint.

NBA Odds Tracker | Python, PostgreSQL

nba-odds.up.railway.app

- Built an idempotent data pipeline that pulls live NBA betting odds into a normalized PostgreSQL schema without duplication across repeated runs.
- Wrote transform logic to convert decimal to American odds and UTC to EST, with clean reinsertion on refresh.

WORK EXPERIENCE

Sales Associate

March 2021 – May 2025

Quiksilver, Wall, NJ

- Handled customer questions and product recommendations in a fast-paced retail environment, often managing multiple requests at once.
- Balanced part-time work with a full CS course load; stayed organized across competing priorities.